ANUSHKA DESAI

23070122035

Project 6:

Social Media underlying infra challenges-Using Kubernetes, complete the features of application scalability by demonstrating how to create a cluster with the autoscale feature.

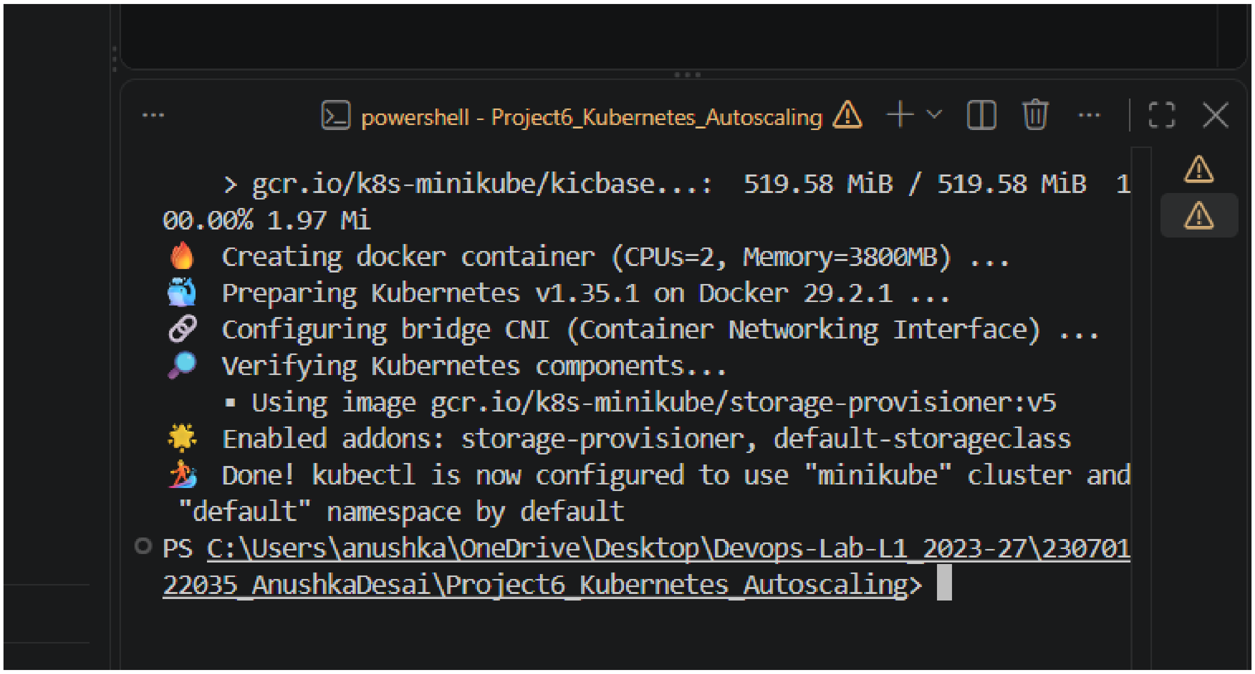
1. Kubernetes, Minikube and Docker CLI Verification

```
PS C:\Users\anushka\OneDrive\Desktop\Devops-Lab-L1_2023-27\230701
22035_AnushkaDesai> minikube version
minikube version: v1.38.1
commit: c93a4cb9311efc66b90d33ea03f75f2c4120e9b0
PS C:\Users\anushka\OneDrive\Desktop\Devops-Lab-L1_2023-27\230701
22035_AnushkaDesai> docker --version
Docker version 29.6.1, build 8900f1d
PS C:\Users\anushka\OneDrive\Desktop\Devops-Lab-L1_2023-27\230701
22035_AnushkaDesai> kubectl version --client
Client Version: v1.36.1
Kustomize Version: v5.8.1
PS C:\Users\anushka\OneDrive\Desktop\Devops-Lab-L1_2023-27\230701
22035_AnushkaDesai>
```

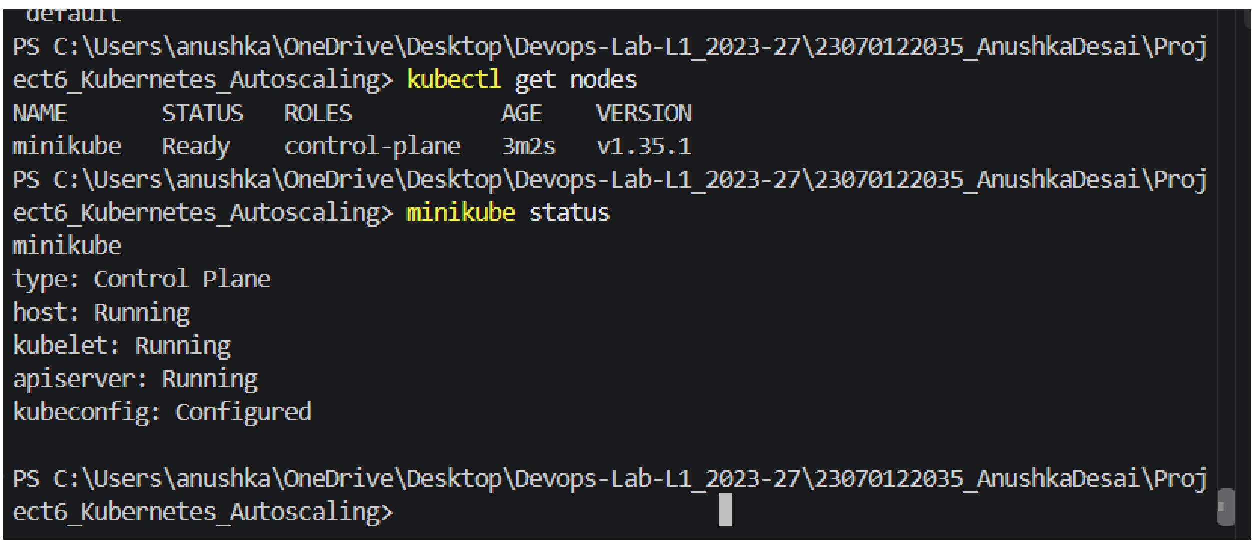
2. Docker Engine Verification

```
PS C:\Users\anushka\OneDrive\Desktop\De
22035_AnushkaDesai> docker info
 OSType: linux
 Architecture: x86_64
 CPUs: 12
 Total Memory: 7.372GiB
 Name: docker-desktop
 ID: 7ad63502-199e-477e-8324-866eda177
 Docker Root Dir: /var/lib/docker
 Debug Mode: false
 HTTP Proxy: http.docker.internal:3128
 HTTPS Proxy: http.docker.internal:312
 No Proxy: hubproxy.docker.internal
 Labels:
  com.docker.desktop.address=npipe://\
 Experimental: false
 Insecure Registries:
  hubproxy.docker.internal:5555
  ::1/128
  127.0.0.0/8
 Live Restore Enabled: false
 Firewall Backend: iptables
```

3. Kubernetes Cluster Creation Using Minikube



4. Kubernetes Cluster and Node Status Verification



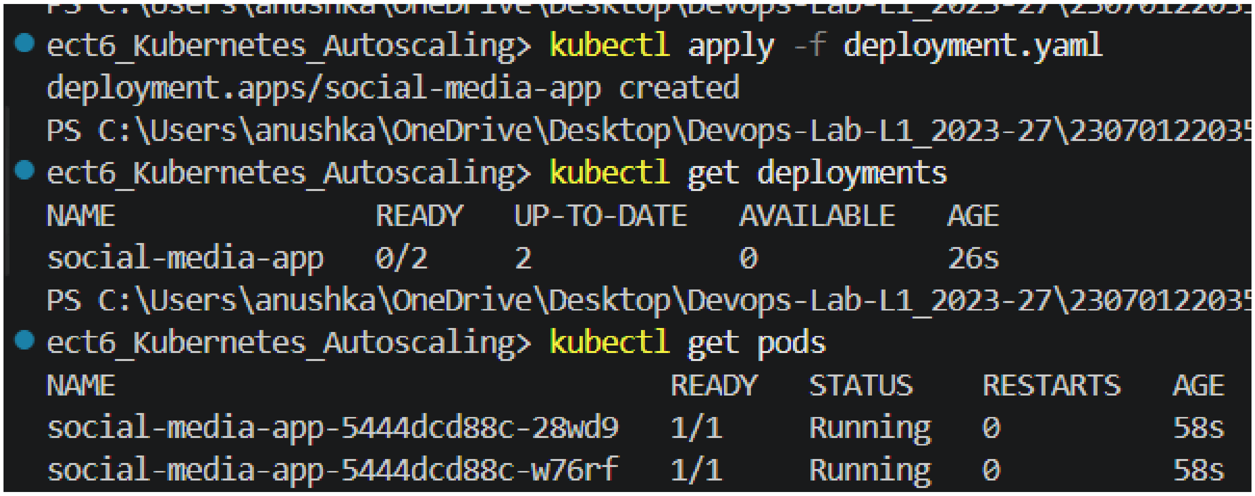
5. Minikube Profile and Kubernetes System Pods

```
PS C:\Users\anushka\OneDrive\Desktop\Devops-Lab-L1_2023-27\23070122035_AnushkaDesai\Proj
ect6_Kubernetes_Autoscaling>                          minikube profile list
┌──────────┬────────┬─────────┬──────────────┬─────────┬────────┬───────┬────────────────
─┬──────────────────┐
│ PROFILE  │ DRIVER │ RUNTIME │      IP      │ VERSION │ STATUS │ NODES │ ACTIVE PROFILE
 │ ACTIVE KUBECONTEXT │
├──────────┼────────┼─────────┼──────────────┼─────────┼────────┼───────┼────────────────
─┼──────────────────┤
│ minikube │ docker │ docker  │ 192.168.49.2 │ v1.35.1 │ OK     │ 1     │ *
 │ *                │
└──────────┴────────┴─────────┴──────────────┴─────────┴────────┴───────┴────────────────
─┴──────────────────┘
PS C:\Users\anushka\OneDrive\Desktop\Devops-Lab-L1_2023-27\23070122035_AnushkaDesai\Proj
ect6_Kubernetes_Autoscaling> kubectl get pods -A
NAMESPACE     NAME                               READY   STATUS    RESTARTS   AGE
kube-system   coredns-7d764666f9-9v2f6           1/1     Running   0          5m52s
kube-system   etcd-minikube                      1/1     Running   0          5m59s
kube-system   kube-apiserver-minikube            1/1     Running   0          6m2s
kube-system   kube-controller-manager-minikube   1/1     Running   0          6m1s
kube-system   kube-proxy-9vwk9                   1/1     Running   0          5m52s
kube-system   kube-scheduler-minikube            1/1     Running   0          6m
kube-system   storage-provisioner                1/1     Running   0          5m46s
PS C:\Users\anushka\OneDrive\Desktop\Devops-Lab-L1_2023-27\23070122035_AnushkaDesai\Proj
ect6_Kubernetes_Autoscaling>
```

6. Metrics Server and Kubernetes Node Resource Metrics

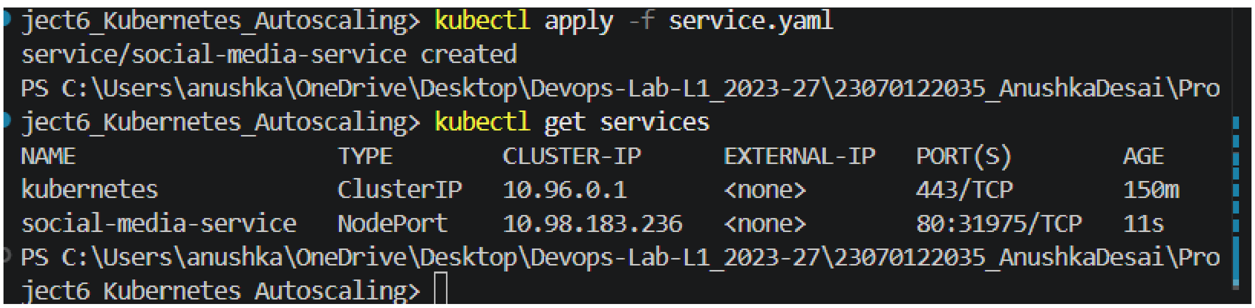
```
PS C:\Users\anushka\OneDrive\Desktop\Devops-Lab-L1_2023-27\23070122035_AnushkaDesai\Proj
ect6_Kubernetes_Autoscaling> kubectl get pods -n kube-system
NAME                               READY   STATUS    RESTARTS   AGE
coredns-7d764666f9-9v2f6           1/1     Running   0          10m
etcd-minikube                      1/1     Running   0          10m
kube-apiserver-minikube            1/1     Running   0          10m
kube-controller-manager-minikube   1/1     Running   0          10m
kube-proxy-9vwk9                   1/1     Running   0          10m
kube-scheduler-minikube            1/1     Running   0          10m
metrics-server-9d74bb658-ndmp6     1/1     Running   0          3m14s
storage-provisioner                1/1     Running   0          10m
PS C:\Users\anushka\OneDrive\Desktop\Devops-Lab-L1_2023-27\23070122035_AnushkaDesai\Proj
ect6_Kubernetes_Autoscaling> kubectl top nodes
NAME       CPU(cores)   CPU(%)   MEMORY(bytes)   MEMORY(%)
minikube   420m         3%       609Mi           8%
```

7. Social Media Application Deployment with Two Replicas



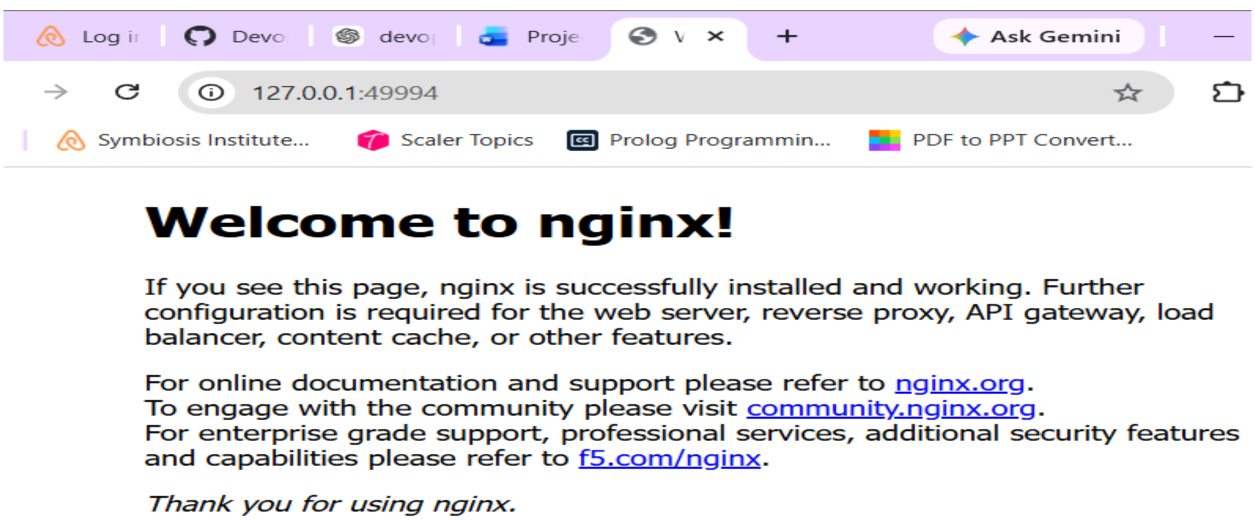
```
PS C:\Users\anushka\OneDrive\Desktop\Devops-Lab-L1_2023-27\2307012203
ect6_Kubernetes_Autoscaling> kubectl apply -f deployment.yaml
deployment.apps/social-media-app created
PS C:\Users\anushka\OneDrive\Desktop\Devops-Lab-L1_2023-27\2307012203
ect6_Kubernetes_Autoscaling> kubectl get deployments
NAME               READY   UP-TO-DATE   AVAILABLE   AGE
social-media-app   0/2     2            0           26s
PS C:\Users\anushka\OneDrive\Desktop\Devops-Lab-L1_2023-27\2307012203
ect6_Kubernetes_Autoscaling> kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
social-media-app-5444dcd88c-28wd9   1/1     Running   0          58s
social-media-app-5444dcd88c-w76rf   1/1     Running   0          58s
```

8. Kubernetes NodePort Service for Social Media Application



```
ject6_Kubernetes_Autoscaling> kubectl apply -f service.yaml
service/social-media-service created
PS C:\Users\anushka\OneDrive\Desktop\Devops-Lab-L1_2023-27\23070122035_AnushkaDesai\Pro
ject6_Kubernetes_Autoscaling> kubectl get services
NAME                   TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)        AGE
kubernetes             ClusterIP   10.96.0.1       <none>        443/TCP        150m
social-media-service   NodePort    10.98.183.236   <none>        80:31975/TCP   11s
PS C:\Users\anushka\OneDrive\Desktop\Devops-Lab-L1_2023-27\23070122035_AnushkaDesai\Pro
ject6_Kubernetes_Autoscaling>
```

9. Social Media Application Running Through Kubernetes Service



10. Horizontal Pod Autoscaler Configuration and CPU Target

```
PS C:\Users\anushka\OneDrive\Dkubectl get hpaab-L1_2023-27\23070122035_AnushkaDesai\Pr
oject6_Kubernetes_Autoscaling>
NAME              REFERENCE                        TARGETS       MINPODS    MAXPODS    REP
LICAS    AGE
social-media-hpa   Deployment/social-media-app    cpu: 0%/50%    2          5          2
```

11. Kubernetes Horizontal Pod Autoscaling Under CPU Load

```
PS C:\Users\anushka\OneDrive\Desktop\Devops-Lab-L1_2023-27\23070122035_AnushkaDesai\Project6_Kubernetes_Autoscalin
g> kubectl get hpa
NAME               REFERENCE                        TARGETS         MINPODS    MAXPODS    REPLICAS    AGE
social-media-hpa   Deployment/social-media-app      cpu: 218%/50%   2          5          4           13m
PS C:\Users\anushka\OneDrive\Desktop\Devops-Lab-L1_2023-27\23070122035_AnushkaDesai\Project6_Kubernetes_Autoscalin
g> kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
social-media-app-5444dcd88c-28wd9   1/1     Running   0          175m
social-media-app-5444dcd88c-5c7f6   1/1     Running   0          31s
social-media-app-5444dcd88c-7j4gd   1/1     Running   0          46s
social-media-app-5444dcd88c-cs2nt   1/1     Running   0          46s
social-media-app-5444dcd88c-w76rf   1/1     Running   0          175m
PS C:\Users\anushka\OneDrive\Desktop\Devops-Lab-L1_2023-27\23070122035_AnushkaDesai\Project6_Kubernetes_Autoscalin
g>
```

12. Horizontal Pod Autoscaler Reaching Maximum Replicas

```
g>                              kubectl get hpa
NAME               REFERENCE                        TARGETS         MINPODS    MAXPODS    REPLICAS    AGE
social-media-hpa   Deployment/social-media-app      cpu: 100%/50%   2          5          5           15m
```

13. Kubernetes Autoscaling After CPU Load Reduction

```
g> kubectl get hpa
NAME               REFERENCE                        TARGETS       MINPODS    MAXPODS    REPLICAS    AGE
social-media-hpa   Deployment/social-media-app      cpu: 0%/50%   2          5          2           26m
PS C:\Users\anushka\OneDrive\Desktop\Devops-Lab-L1_2023-27\23070122035_AnushkaDesai\Project6_Kuber
g> kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
social-media-app-5444dcd88c-28wd9   1/1     Running   0          3h8m
social-media-app-5444dcd88c-w76rf   1/1     Running   0          3h8m
```